

1. Arithmetic Sequences & Series

Term	Formula
n-th term (first term a , common difference d)	$T_n = a + (n - 1)d$
Sum of first n terms	$S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{n}{2}(a + T_n)$
Sum of consecutive integers 1..n	$\sum_{i=1}^n i = \frac{n(n+1)}{2}$
Sum of consecutive odd numbers 1+3+...+(2n-1)	n^2
Sum of consecutive even numbers 2+4+...+2n	$n(n + 1)$

2. Geometric Sequences & Series

Term	Formula
n-th term (first term a , common ratio r)	$T_n = a r^{n-1}$
Sum of first n terms ($r \neq 1$)	$S_n = a \cdot \frac{r^n - 1}{r - 1}$
Sum of infinite geometric series ($\ r\ < 1$)	$S_\infty = \frac{a}{1-r}$

3. Sums of Powers (Faulhaber's formulae)

Sum	Formula
$\sum_{i=1}^n i$	$\frac{n(n+1)}{2}$
$\sum_{i=1}^n i^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{i=1}^n i^3$	$\left(\frac{n(n+1)}{2}\right)^2$

4. Combinatorics

Concept	Formula
Factorial	$n! = 1 \cdot 2 \cdots n$
Binomial coefficient (n choose k)	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$
Permutations (nPr)	$P(n, k) = \frac{n!}{(n-k)!}$
Stars and bars (non-negative solutions)	$\binom{n+k-1}{k-1}$
Catalan numbers	$C_n = \frac{1}{n+1} \binom{2n}{n}$
Bell numbers (partitions)	$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$
Fibonacci numbers	$F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$
Binomial theorem	$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$
Vandermonde's identity	$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$



5. Modular Arithmetic

Operation	Formula / Rule
Addition	$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$
Multiplication	$(a \cdot b) \bmod m = ((a \bmod m) \cdot (b \bmod m)) \bmod m$
Modular inverse (if $\gcd(a,m)=1$)	$a^{-1} \equiv a^{\varphi(m)-1} \pmod{m}$ (Euler) $a^{-1} \equiv a^{m-2} \pmod{m}$ (Fermat, if m prime)
Fermat's little theorem (p prime)	$a^{p-1} \equiv 1 \pmod{p}$
Euler's totient	$\varphi(n) = n \prod_{p n} \left(1 - \frac{1}{p}\right)$
Chinese remainder theorem	System $x \equiv a_i \pmod{m_i}$ with pairwise coprime moduli has a unique solution modulo $\prod m_i$.

6. Number Theory

Concept	Formula
GCD	$\gcd(a, b) = \gcd(b, a \bmod b)$
LCM	$\text{lcm}(a, b) = \frac{a \cdot b}{\gcd(a, b)}$
Sieve of Eratosthenes	Mark multiples, $O(n \log \log n)$
Prime counting	$\pi(n) \approx n / \ln n$ (asymptotic)
Divisor count	If $n = \prod p_i^{e_i}$, number of divisors = $\prod (e_i + 1)$
Sum of divisors	$\sigma(n) = \prod \frac{p_i^{e_i+1} - 1}{p_i - 1}$



7. Bitwise Identities (Bonus)

Identity	Formula
OR + AND = Sum	$x \mid y + x \& y = x + y$
XOR = OR - AND	$x \oplus y = (x \mid y) - (x \& y)$
XOR = Sum - 2·AND	$x \oplus y = x + y - 2(x \& y)$
AND = (OR - XOR)	$x \& y = (x \mid y) - (x \oplus y)$
AND distributes over XOR	$x \& (y \oplus z) = (x \& y) \oplus (x \& z)$
De Morgan	$\sim (x \mid y) = (\sim x) \& (\sim y)$
Modulo by power of two	$x \bmod 2^k = x \& (2^k - 1)$
Clear lowest set bit	$x \& (x-1)$
Isolate lowest set bit	$x \& -x$
Set all bits below lowest zero	$x \mid (x-1)$

8. Miscellaneous

Concept	Formula
Harmonic numbers	$H_n = \sum_{i=1}^n \frac{1}{i} \approx \ln n + \gamma$
Stirling numbers of the second kind	$S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$
Catalan's convolution	$\sum_{i=0}^n C_i C_{n-i} = C_{n+1}$



1. Basic addition/subtraction identities

$x + y = (x | y) + (x \& y)$
(because $\text{sum} = \text{OR} + \text{AND}$, no carries overlap)

$x | y = x + y - (x \& y)$
(OR = sum - AND)

$x \wedge y = (x | y) - (x \& y) = x + y - 2(x \& y)$
(XOR = OR - AND, and also = sum - 2·AND)

$x \wedge y = (x \& \sim y) | (\sim x \& y)$
(XOR in terms of AND/NOT)

2. Shifts and powers of two

$x \ll k = x * 2^k$ (left shift = multiplication by power of two)

$x \gg k = x / 2^k$ for non-negative `x` (right shift = floor division)

$x \bmod 2^k = x \& (2^k - 1)$ (modulo power of two = bitmask of lower k bits)

3. Clearing / toggling / extracting bits

$x \& (x - 1) \rightarrow$ clears the lowest set bit of `x`.

$x \& -x \rightarrow$ isolates the lowest set bit (only that bit remains).

$x | (x - 1) \rightarrow$ sets all lower bits to 1 up to the lowest zero.

$x \wedge (x - 1) \rightarrow$ gives a mask of trailing ones (e.g., `...0111`).

$(x \& y) == 0$ means `x` and `y` have no overlapping set bits (disjoint).

4. De Morgan's laws (bitwise)

$\sim(x | y) = (\sim x) \& (\sim y)$

$\sim(x \& y) = (\sim x) | (\sim y)$

5. Relations between AND, OR, XOR

$$x \& y = (x | y) - (x \wedge y)$$

(AND = OR - XOR)

$$x | y = (x \wedge y) + (x \& y)$$

(OR = XOR + AND)

$$x \wedge y = (x | y) \& \sim(x \& y)$$

(XOR = OR and NOT AND)

$$x \wedge y = (x \& \sim y) | (\sim x \& y)$$

(set bits in exactly one operand)

6. Identities with subtraction and borrows

$$x - y = (x \wedge y) - 2 * (\sim x \& y)$$

(for two's complement, but often not used)

$$x - y = (x \& \sim y) - (\sim x \& y)$$

(difference = bits in x not y minus bits in y not x)

7. Symmetry and distributivity

AND, OR, XOR are commutative and associative.

****AND distributes over OR**** and vice versa (bitwise distributive laws):

$$x \& (y | z) = (x \& y) | (x \& z)$$

$$x | (y \& z) = (x | y) \& (x | z)$$

- ****XOR is associative****, but does ****not**** distribute over AND/OR (though $x \& (y \wedge z) \neq (x \& y) \wedge (x \& z)$ is actually false; actually it ****does**** distribute: $x \& (y \wedge z) = (x \& y) \wedge (x \& z)$ – correct: AND distributes over XOR).

8. Multiplicative relations (with carries)

- ****Carry-less multiplication**** (used in GF(2)):

$$x \wedge y \& z = (x \& z) \wedge (y \& z) \text{ (AND distributes over XOR)}$$

- ****No simple identity**** for $(x | y) * z$ without carries; bitwise ops and multiplication are generally “incompatible” unless restricted to powers of two.